Supplementary Material

Neighbours and Orientation: Out-of-Distribution Detection with Frozen Diffusion Features

Anonymous Submission

Section and table numbers of the form Sn refer to this document; plain numbers refer to the paper. The anonymised code, the protocol documents written before each evaluation, and the per-image scores behind every table are provided with this submission (Section S17).

S1 Protocol and Data

Splits. For every ID dataset the images are split into three disjoint parts. (i) *Selection data*: ID development images and OOD datasets that are never used for evaluation (for the five 32×32 ID datasets: STL-10, GTSRB, Flowers-102, EuroSAT, MNIST in RGB, Oxford-IIIT Pets, Caltech-101, Food-101, FGVC-Aircraft, Imagenette and Rendered SST-2; for ImageNet-64 the same without Imagenette; for FashionMNIST: EMNIST letters, Kuzushiji-49, Semeion and greyscale CIFAR-10, SVHN and GTSRB). (ii) *Development evaluation images*: 1,000 per set (2,000 for FashionMNIST). (iii) *Final evaluation images*: 500 per set. The configuration of Upright kNN (block set and R of the kNN term) was chosen on (i) among 20 candidates (half-resolution block or the mean of the three blocks; R among ten values from 1 to 1000) by the mean AUROC of the term alone; the head (inputs, linear vs. MLP, temperature) was chosen on held-out bank images only. The form of the score was developed on (ii), which draws on the same OOD datasets as (iii); a written protocol was then frozen and (iii) was evaluated once. All tables of the paper and of this document are on (iii) except the development ablations (Table S13), the weight sweep (Table S4, which also reports (iii)) and the mirror test (Table S17).

OOD sets. For the 32×32 ID datasets: the other natural datasets among CIFAR-10/100, SVHN and CelebA, plus Textures (DTD), Tiny-ImageNet (validation images resized to 32×32), the LSUN-FIX and ImageNet-FIX sets distributed by Tack *et al.* [2020] (the corrected resizing, without the artefacts of the original resized LSUN/ImageNet sets), ImageNet32 (downsampled validation images) and Places365 (validation images). For ImageNet-64: the OpenOOD v1.5 sets (SSB-hard, NINCO, iNaturalist, Textures, OpenImage-O) plus Places365 and CelebA. For FashionMNIST: MNIST, KMNIST, Omniglot and notMNIST embedded in 32×32. Every ID dataset is also paired with its own images flipped upside down and, except ImageNet-64, with uniform-noise and constant images, giving 63 pairs.

Bank and noise. The bank is 50,000 ID training images (10,000 for ImageNet-64, 10 per class). The noise image ε of $\sigma_1\varepsilon$ is one fixed draw per noise seed, shared by all images; the results are averaged over three seeds, and the head of each seed is trained on the features of that seed with a fixed initialisation seed.

S2 Models and Configuration

ID	Res.	Model	Source	d	R	T	Head acc. (%)	tuned kNN configuration
CIFAR-10	32×32	EDM (VP, DDPM++)	[Karras *et al.*, 2022]	19.8	2	1.43	95.5	mean of 3 blocks, σ_1, mean of top 5
SVHN	32×32	VP-SDE	[Kamkari *et al.*, 2024]	22.8	5	1.43	86.7	mean of 3 blocks, σ_1, mean of top 10
CelebA	32×32	iDDPM	[Nichol and Dhariwal, 2021]	25.1	5	0.72	100.0	mean of 3 blocks, σ_7, mean of top 5
CIFAR-100	32×32	DDPM++ (EDM precond.)	[Chen *et al.*, 2024]	24.0	5	1.38	88.2	mean of 3 blocks, σ_1, mean of top 10
ImageNet-64	64×64	EDM2-XS (uncond.)	[Karras *et al.*, 2024]	20.2	1000	1.18	77.3	mean of 3 blocks, σ_1, 1000th neighbour
FashionMNIST	32×32	U-Net (EDM precond.)	trained by us	14.4	1	0.97	99.7	half-res. block, σ_7, mean of top 2

Table S1: ID datasets, diffusion models and the configuration of Upright kNN (noise seed 0). All models are unconditional; all weights except FashionMNIST are public and unchanged. d: Levina–Bickel intrinsic dimension of the bank features (20 neighbours, 3,000 bank points); R: neighbour rank chosen on the selection data; T: temperature of the head; head accuracy: rotation accuracy on the 5,000 held-out bank images. The last column gives the configuration of the tuned kNN baseline (chosen on the selection data among 80; σ_7 is the seventh of the model's noise levels in increasing order).

Features. The three hooked modules are the last decoder blocks at the full, half and quarter resolution of the network input (for DiT-XL/2, which has no resolutions, blocks 28, 21 and 14 of 28), read after the block's residual sum. For each block with C channels, $\mathbf{g}$ is the channel-wise mean and standard deviation over the spatial grid ($2C$ values) and $\mathbf{q}$ the same over each cell of a 2×2 grid ($8C$ values); the head reads $[\mathbf{g}, \mathbf{q}]$ of the three blocks (for example 7,680 values for EDM2-XS), standardised per feature with the mean and standard deviation over the four rotated views of the bank. **FashionMNIST U-Net.** An EDM-style U-Net (39M parameters, the architecture of our chest X-ray and histology models below) trained with the EDM preconditioning and loss on 55,000 training images for 8M images seen, batch 128, learning rate $2.5 \cdot 10^{-4}$, dropout 0.13, without augmentation; the 5,000 remaining training images are held out. **Head.** MLP with one hidden layer of 512 ReLU units, AdamW (learning rate 10^{-3}, weight decay 10^{-4}) with a one-cycle schedule over 8 epochs, batches of 512 bank images times four rotations, 5,000 held-out bank images for the temperature (200 values of T on a logarithmic grid from 0.25 to 8, minimum held-out negative log-likelihood); a linear head is trained first for reference only. **Latent models.** For EDM2-XS at 512 pixels and DiT-XL/2 the noise is added to the latents and the bank images are rotated in pixel space before encoding; DiT-XL/2 is class-conditional and is queried with its null class (the unconditional branch of classifier-free guidance).

What in the kNN rule carries its gain over CFS (kNN). Table S2 separates the features from the head on the final images (a post hoc reading of frozen scores; nothing was chosen from it; the development images give the same picture within 0.3 points). CFS (kNN) scores its descriptors by the mean Euclidean distance to the 10 nearest bank descriptors after standardising each dimension; our kNN rule takes the R-th largest cosine of the unit-normalised statistics. Swapping the head alone, on CFS's own descriptors, gives 2.3 points (85.5 to 87.8%): the cosine of unit vectors is the better metric for these pooled statistics on every dataset that is not saturated. Swapping the block instead (our half-resolution decoder block for CFS's encoder and decoder blocks at 16×16, both under the cosine head) changes the mean by -0.1 points but trades datasets, +7.0 on CIFAR-100 against -10.3 on ImageNet-64; the statistic (10th largest or mean of the 10 largest) matters little (+0.1). Table S3 then walks the 80 configurations among which the tuned kNN was chosen. Pooling the three decoder blocks adds 1.2 points (+3.6 on CIFAR-10, +4.4 on CIFAR-100, -1.5 on ImageNet-64; the near-OOD mean from 60.5 to 64.6%). The higher noise level σ_7 costs -1.9 points on average (CIFAR-100 from 80.3 to 61.6%) while helping ImageNet-64 at small R (59.1 to 67.0%). The neighbour rank R is what the datasets disagree about: ImageNet-64 wants $R = 1{,}000$, a tenth of its bank (57.6% at $R = 10$, 77.0% at $R = 1{,}000$), whereas CIFAR-10, CIFAR-100 and FashionMNIST peak at $R \leq 2$. The best single configuration reaches 90.2% (three blocks, σ_1, mean of the 1,000 largest cosines); choosing R and the statistic per dataset on the selection data gives the tuned kNN's 92.6%, 3.6 points over the three blocks at $R = 10$. In sum, of the 7.1 points between CFS (kNN) and the tuned kNN, 2.3 are the metric, 1.2 the three blocks and 3.6 the per-dataset neighbour rank; the choice of block and of noise level adds nothing on average.

Features and head	CIFAR-10	SVHN	CelebA	CIFAR-100	ImageNet-64	FMNIST	Mean	Near	Flip
CFS descriptors (encoder + decoder 16×16 *blocks, logSNR 7)*									
standardised Euclidean, mean of 10 (CFS)	84.9	92.1	99.9	72.3	63.9	99.9	85.5	62.8	70.6
standardised Euclidean, 10th	83.3	91.0	99.9	70.6	64.8	99.9	84.9	62.4	69.5
cosine, 10th (ours)	88.5	95.6	99.9	73.3	69.4	100.0	87.8	65.1	72.5
cosine, mean of 10	89.6	95.6	100.0	74.4	68.8	100.0	88.1	65.5	73.3
our half-resolution decoder block (σ_1)									
standardised Euclidean, mean of 10 (CFS)	86.2	96.7	99.9	78.6	61.9	99.6	87.2	59.7	67.2
standardised Euclidean, 10th	85.1	96.7	99.9	77.5	64.0	99.5	87.1	59.5	66.5
cosine, 10th (ours)	88.6	98.7	100.0	80.3	59.1	99.7	87.7	60.5	68.2
cosine, mean of 10	89.5	98.7	100.0	81.2	58.0	99.7	87.9	60.8	68.7

Table S2: Features and head of the kNN rule (final images; AUROC, %, mean over the natural OOD sets of each ID dataset, three noise seeds; Near: the four near-OOD pairs; Flip: the upside-down probe). The CFS descriptors and our half-resolution block, each scored with the CFS head (standardised Euclidean distance, mean of the 10 nearest bank images), with ours (cosine of the unit-normalised statistics, 10th largest) and with the two mixed statistics; bank of 50,000 ID training images (ImageNet-64: 10,000). The first row is CFS (kNN) of Table 2 and the seventh the basic kNN.

			R									
Statistic	Blocks	Noise	1	2	5	10	20	50	100	200	500	1000
R-th largest cosine	half-res. block	σ_1	87.7	87.8	87.7	87.7	87.8	87.9	88.1	88.3	88.6	88.5
R-th largest cosine	half-res. block	σ_7	86.2	86.1	85.9	85.8	85.6	85.3	85.0	84.6	83.9	83.4
R-th largest cosine	three blocks	σ_1	88.9	88.9	88.9	88.9	88.9	89.0	89.2	89.5	90.1	90.1
R-th largest cosine	three blocks	σ_7	87.2	87.1	86.9	86.6	86.3	85.9	85.5	85.0	84.2	83.5
mean of the R largest	half-res. block	σ_1	87.7	87.9	87.9	87.9	87.8	87.9	88.0	88.1	88.6	88.7
mean of the R largest	half-res. block	σ_7	86.2	86.4	86.3	86.1	85.9	85.7	85.4	85.1	84.6	84.1
mean of the R largest	three blocks	σ_1	88.9	89.0	89.1	89.0	89.0	89.0	89.1	89.3	89.8	**90.2**
mean of the R largest	three blocks	σ_7	87.2	87.4	87.2	87.0	86.8	86.4	86.1	85.7	85.0	84.4

Table S3: The 80 configurations of the tuned kNN (final images; six-dataset mean AUROC over the natural OOD sets, three noise seeds). The tuned kNN of Table 2 picks one configuration per ID dataset on the selection data (Table S1); the basic kNN is the R-th largest cosine of the half-resolution block at σ_1 with $R = 10$. Bold: the best single configuration.

S3 Comparison Methods

All likelihood- and statistic-based methods use the same diffusion models. NLL, IC and TT use the probability-flow ODE with 64 Heun steps (128 forward and 128 backward evaluations). MSMA* is the Mahalanobis distance of the vector of score norms at 20 noise levels, fitted on ID development images. DiffPath and GEPC follow the official code and EigenScore the paper's procedure. DDPM-OOD [Graham *et al.*, 2023] reconstructs each image from 25 noise levels with 50 Heun steps each (2,475 evaluations), and averages the MSE and LPIPS errors z-scored with 128 ID development images, as in the official code. LMD [Liu *et al.*, 2023] inpaints each image 10 times with alternating 8×8 checkerboard masks by the 1,000-step predictor–corrector sampler of the official code and scores the median LPIPS distance. The likelihood + LID rule of Kamkari *et al.* [2024] estimates the local intrinsic dimension from the singular values of $4D$ score evaluations at the VP time $t_0 = 0.01$ (the public configuration), with the singular-value threshold calibrated on 128 training images to the model-free LPCA estimate of the public configuration, and combines it with the probability-flow ODE (PF-ODE) likelihood; its AUROC is the area under the Pareto frontier of the ROC points of all threshold pairs (the public `roc_analysis`), an oracle over threshold pairs, so it is optimistic. The public configurations cover the five 32×32 ID datasets, so the rule is not run on ImageNet-64. CFS follows the official code (hook policy: up to four encoder and four decoder candidates in structural order, the one with the largest variance on 32 ID training images selected in each region; channel mean and standard deviation; diagonal-Gaussian and standardised 10-NN heads fitted on the bank). At the paper's default noise level (logSNR 5) our port reproduces the paper's CIFAR-10 numbers on the four shared pairs within 0.5 points; the official configuration (logSNR 7) is better on the development images and is used throughout. The rotation classifier is a ResNet-18 trained from scratch for 30 epochs on the ID training images; its score is the sum of the log-probabilities of the correct angle over the four rotated inputs. CSI is described in Section S15. CPU times use a randomly initialised CIFAR-10 network (the cost

does not depend on the weights) on an Intel Xeon (2.30 GHz, 2 threads), every row in one session: the multi-pass methods once, the single-pass methods interleaved (median of 15 rounds), so that run-to-run drift does not land on one method; the kNN head of CFS is timed with the bank norms precomputed (`torch.cdist`, which recomputes them at every call, takes 0.19 s for the same scores). The single-pass detectors all cost about one denoiser pass (kNN 0.16 s, CFS 0.16 and 0.16 s, Upright kNN 0.17 s); the CPU times of DDPM-OOD, LMD, LID+NLL and CSI were not measured, their NFE column gives the scale.

S4 Proof of Proposition 2

For $\mathbf{G} \in \mathcal{G}$ the horizontal flip satisfies $\mathbf{GH} = \mathbf{HG}^{-1}$ (a flip reverses the sense of rotation). With $\mathbf{V} = \mathbf{HR}_{180}$ and $p(\mathbf{Hx}) = p(\mathbf{x})$ for all $\mathbf{x}$,

$$\bar{p}(\mathbf{Vx}) = \frac{1}{4}\sum_{\mathbf{G}\in\mathcal{G}} p(\mathbf{GHR}_{180}\mathbf{x}) = \frac{1}{4}\sum_{\mathbf{G}\in\mathcal{G}} p(\mathbf{HG}^{-1}\mathbf{R}_{180}\mathbf{x}) = \frac{1}{4}\sum_{\mathbf{G}\in\mathcal{G}} p(\mathbf{G}^{-1}\mathbf{R}_{180}\mathbf{x}) = \bar{p}(\mathbf{x}), \tag{1}$$

since $\mathbf{G}^{-1}\mathbf{R}_{180}$ runs over $\mathcal{G}$ exactly once as $\mathbf{G}$ does. Subtracting Proposition 1 at $\mathbf{x}$ from Proposition 1 at $\mathbf{Vx}$ gives $\log p(\mathbf{Vx}) - \log p(\mathbf{x}) = \log P(\text{up} \mid \mathbf{Vx}) - \log P(\text{up} \mid \mathbf{x})$. □

S5 The Weight of the Two Terms

w	0	0.25	0.5	0.75	1	1.25	1.5	2	3	4
development images										
mean, natural OOD sets	92.44	92.84	93.08	93.23	93.31	93.34	93.36	93.33	93.19	93.04
mean, near-OOD pairs	69.98	70.38	70.62	70.73	70.75	70.74	70.69	70.55	70.22	69.90
mean, upside-down probe	72.57	78.26	81.97	84.56	86.39	87.72	88.69	90.00	91.39	92.11
CIFAR-10	93.9	94.9	95.4	95.7	95.8	95.9	96.0	96.0	96.0	95.9
SVHN	99.1	99.3	99.4	99.4	99.4	99.3	99.3	99.2	98.9	98.6
CelebA	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
CIFAR-100	83.6	84.0	84.3	84.4	84.5	84.5	84.5	84.4	84.1	83.7
ImageNet-64	78.1	78.9	79.5	79.9	80.2	80.4	80.4	80.5	80.3	80.1
FashionMNIST	99.9	99.9	99.9	99.9	99.9	99.9	99.9	99.9	99.9	99.9
final images										
mean, natural OOD sets	92.50	92.90	93.13	93.27	93.35	93.40	93.43	93.42	93.31	93.17
mean, near-OOD pairs	69.94	70.33	70.59	70.77	70.86	70.90	70.90	70.85	70.66	70.42
mean, upside-down probe	72.84	78.46	82.20	84.81	86.64	87.95	88.93	90.27	91.68	92.40
CIFAR-10	93.7	94.6	95.0	95.3	95.4	95.4	95.4	95.4	95.4	95.3
SVHN	99.4	99.6	99.7	99.7	99.7	99.7	99.7	99.6	99.3	99.0
CelebA	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
CIFAR-100	85.0	85.2	85.4	85.5	85.5	85.5	85.4	85.3	84.9	84.4
ImageNet-64	77.0	78.0	78.7	79.2	79.6	79.8	80.0	80.3	80.4	80.3
FashionMNIST	99.9	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Table S4: Mean AUROC (%) of $s_w = -d \log r_R + w \log q_\phi(\mathbf{R}_0 \mid \mathbf{x})$ for several relative weights w (noise seeds 0–2); $w = 1$ is Upright kNN, $w = 0$ its kNN term. Because the kNN term is $-d \log r_R$, another estimate d' of the intrinsic dimension corresponds to $w = d/d'$: a 30% error in d moves w between 0.77 and 1.43, inside the plateau, and the estimates with 5 or 50 neighbours (Table S8) correspond to $w \approx 0.75$ and 1.1. The near-OOD pairs are CIFAR-10↔CIFAR-100, SSB-hard and NINCO. This analysis was added after the frozen evaluation; the paper's score was fixed at $w = 1$ before any of these numbers was computed.

S6 Two Notions: Details and Robustness

Analysis. All quantities of Table 1 of the paper are computed from the stored features of the frozen method (no change to it). *Geometry*: in the space of the kNN term (unit-length channel mean and standard deviation of the three blocks), the distance between each of 4,000 random bank images and its own half turn, divided by its distance to its R-th nearest

other bank image (R of the frozen protocol); median. *Rotation share*: the between-rotation variance (variance of the four rotation means of the standardised head-input features) divided by the total variance, an estimator-free quantity. *Discriminant and principal directions*: in the head's input space, a linear discriminant analysis (LDA) of the four rotations of the bank images (within-class scatter regularised by a ridge λ), the top principal directions of all rotated views, and 20 draws of random orthonormal directions; we report the variance carried by three directions relative to three random directions. *Projected kNN*: the 10th-nearest-neighbour distance (Euclidean; cosine as a variant) to 10,000 bank images in the projection onto those directions, as a score for ID test images against upside-down ID images and against every natural OOD set. Table 1 uses the fit of the first analysis we ran (4,000 bank images, $\lambda = 10^{-3}$) on the final evaluation images, averaged over the three noise seeds; the largest standard deviation over seeds of a frame-3 natural-OOD mean is 0.6 points.

	LDA fit (final evaluation images, seed 0): var. frame / frame-3 flip / frame-3 natural												development images (seeds 0–2)			
	4k, $\lambda = 10^{-3}$			4k, $\lambda = 10^{-1}$			10k, $\lambda = 10^{-3}$			10k, $\lambda = 10^{-1}$			$\|\Delta_{180}\|/r_R$	fr.-3 flip	fr.-3 nat.	PC-3 nat.
CIFAR-10	0.17	94.5	84.6	2.9	96.5	88.1	0.31	97.6	89.1	4.0	97.7	89.7	1.11	94.1	84.7	75.2
SVHN	0.14	60.3	93.4	3.8	61.9	95.2	0.20	64.3	96.4	4.4	65.3	96.9	1.06	59.7	93.6	89.7
CelebA	1.7	100.0	100.0	20	100.0	100.0	2.3	100.0	100.0	22	100.0	100.0	1.71	100.0	100.0	100.0
CIFAR-100	0.08	74.9	63.1	1.3	78.7	59.8	0.13	80.6	62.0	1.7	80.6	59.6	0.81	75.2	62.4	65.3
ImageNet-64	0.06	78.8	55.6	1.3	83.1	53.5	0.11	87.8	57.3	1.9	86.1	52.2	0.30	78.7	55.3	80.3
FMNIST	0.39	99.9	99.6	7.5	99.9	99.5	0.49	100.0	99.6	8.2	99.9	99.6	2.55	99.9	99.3	93.9

Table S5: Robustness of the two-notions analysis. Left: the LDA fitted on 4,000 or 10,000 bank images with ridge 10^{-3} or 10^{-1}: variance of the three discriminant directions relative to three random directions, and AUROC (%) of the frame-3 kNN on upside-down ID images and on natural OOD sets (mean). The variance share of the discriminant directions depends on the ridge (a larger ridge admits louder directions), the flip AUROC much less. Right: the same analysis on the development evaluation images.

Natural OOD sets in the projections. The paper uses the projected kNN rules only for upside-down images. On natural OOD sets the result depends on the distance (Table S6): with the cosine distance, used in our first (development) analysis, the three discriminant directions beat the top three principal directions on all six ID datasets; with the Euclidean distance they do so on four, and on ImageNet-64 they are at the level of random directions. Either way they do not exceed the orientation term, which reads the frame information out better than a distance in a linear projection does.

	frame-3		PC-3		random 3		kNN	orient.
ID dataset	Eucl.	cos.	Eucl.	cos.	Eucl.	cos.	term	term
CIFAR-10	84.3	89.3	73.8	75.2	55.4	60.1	93.7	93.8
SVHN	93.1	94.3	90.3	85.6	71.7	58.2	99.4	95.7
CelebA	100.0	100.0	100.0	99.9	78.3	78.9	100.0	100.0
CIFAR-100	63.0	68.7	66.5	61.7	58.3	53.5	85.0	72.9
ImageNet-64	55.1	69.5	80.6	64.9	58.5	52.4	77.0	77.4
FMNIST	99.6	99.5	93.2	90.6	66.9	70.8	99.9	99.8

Table S6: Mean AUROC (%) over the natural OOD sets of kNN rules restricted to three directions (final evaluation images, seeds 0–2, fit of Table 1 of the paper; random: mean of 20 draws), with the two terms of Upright kNN for reference.

Controls (`PROTOCOL_E6.md`, written before the results). Five further analyses on the same stored features (Table S7; final evaluation images, three noise seeds): (A) a 10-NN rule in the *full* standardised head space, which sees the 2×2 grid (Euclidean, 10,000 bank images); (B) the estimator-free *mean-3* subspace spanned by the three differences between the mean features of the four rotations (no discriminant analysis, no ridge): its share of the total variance, relative to three random directions, and the 10-NN rule restricted to it; (C) the paper's head trained on the whole-image statistics $\mathbf{g}$ only, i.e. on the input of the kNN term, with the same recipe; (D) the same head on raw-pixel statistics (per channel the mean and standard deviation over the image, the 2×2 grid and the 4×4 grid, 126 values), trained on the rotated bank images; (E) the intrinsic dimension d for other numbers of neighbours and bank points (Table S8). Findings: on the four natural-image datasets a Euclidean distance in the full head space is at chance on upside-down images (47.4–50.4%) and, with per-feature standardisation, weak on most natural OOD sets (the thousands of standardised low-variance features swamp the distance); the cosine distance sees some of the frame

(56.4–81.0%), still far below the linear read-out; on the two aligned datasets (CelebA, FashionMNIST) either distance sees the flip (92.8–95.9%), because a flipped face or garment is far from every bank image. The mean-3 subspace is loud (4.9–20.8% of the total variance) but the rotation shift along it is small compared with the within-rotation spread, so on the natural-image datasets a distance restricted to it barely detects flips (52.2–77.4%); the head on $\mathbf{g}$ alone recovers the frame from the same statistics on which the kNN term is near chance, and is within 3.1 points (flip) and 2.3 points (natural OOD) of the head on $[\mathbf{g}, \mathbf{q}]$, so the grid statistics add little (added to the frozen kNN term, the head on $\mathbf{g}$ alone gives the same mean AUROC on natural OOD sets as Upright kNN; $\mathbf{q}$ is kept because the protocol was frozen with it); the pixel head recovers the frame only where the images are aligned (91.3–96.4% accuracy on CelebA and FashionMNIST against 43.9–64.0% elsewhere) and detects little natural OOD content on the natural-image datasets (50.6–81.9%); and d moves by -11% to $+39\%$ with the number of neighbours, within the plateau of Table S4.

	(A) full head space			(B) mean-3 subspace			(C) head on $\mathbf{g}$ only				(D) head on pixel statistics			
ID dataset	flip	flip (cos.)	natural	var./random	flip	natural	acc.	flip	natural	+kNN term	acc.	flip	natural	+kNN term
CIFAR-10	47.4	81.0	75.3	216	77.4	83.0	92.9	99.0	94.5	95.7	62.3	81.7	63.7	93.4
SVHN	49.0	56.4	99.5	184	52.2	83.4	83.0	81.7	96.6	99.8	64.0	58.8	81.9	98.9
CelebA	92.8	99.6	99.7	436	99.9	99.9	99.8	100.0	100.0	100.0	96.4	99.8	96.5	100.0
CIFAR-100	50.4	60.4	76.3	186	55.8	56.5	83.6	93.8	70.5	85.4	54.6	75.2	57.8	84.7
ImageNet-64	47.6	57.9	48.4	135	55.4	53.3	73.1	94.0	76.9	79.7	43.9	65.4	50.6	76.4
FashionMNIST	95.9	97.3	99.6	265	95.4	97.6	99.4	100.0	99.9	100.0	91.3	95.4	89.1	99.9

Table S7: Controls of the two-notions analysis (AUROC, %, mean over three noise seeds; flip: ID vs upside-down ID; natural: mean over the natural OOD sets; acc.: held-out rotation accuracy; +kNN term: natural-OOD AUROC of the head's orientation term added to the frozen kNN term, to compare with Upright kNN in Table 2 of the paper). (A) and (B) use the Euclidean 10-NN rule after per-feature standardisation; flip (cos.): the same with the cosine distance (the other cosine variants are in the per-seed files).

	3,000 bank points				10,000 bank points			
ID dataset	$k = 5$	$k = 10$	$k = 20$	$k = 50$	$k = 5$	$k = 10$	$k = 20$	$k = 50$
CIFAR-10	26.6	21.7	19.7	18.1	26.3	21.5	19.6	18.0
SVHN	29.6	24.5	22.5	20.8	29.4	24.4	22.5	20.7
CelebA	32.1	26.9	24.8	23.1	32.6	27.1	24.9	23.1
CIFAR-100	31.7	26.3	23.9	21.8	31.9	26.2	23.8	21.8
ImageNet-64	28.0	22.7	20.2	17.9	27.8	22.5	20.0	17.8
FashionMNIST	18.5	15.1	13.4	12.0	18.5	15.0	13.4	12.0

Table S8: Levina–Bickel intrinsic dimension d of the bank features in the kNN space for several numbers of neighbours k and bank points (noise seeds 0–2); the paper uses $k = 20$ on 3,000 points (another random subsample of the bank than the frozen run's, whose values are in Table S1; they agree within 0.3 except FashionMNIST, 13.4 here against 14.4). A different d rescales the kNN term ($w = d/d'$ in Table S4), which changes the mean AUROC by at most 0.13 points over this range.

Further controls (`PROTOCOL_E8.md`, written before the results). Three more analyses on the same stored features and the frozen heads (Table S9; final evaluation images, three noise seeds). (F) *What a low upright posterior looks like.* For the natural OOD images flagged by the orientation term only, by the kNN term only, by both and by neither (thresholds keeping 95% of the final ID images, set on those same images), the shape of the head's posterior over the four angles: *flat* (no angle above 0.5), peaked on a *wrong* angle (maximum ≥ 0.5 at $90°$, $180°$ or $270°$) or peaked on *upright*; the same for ID images and upside-down ID images. (G) The displacement ratio $\|\Delta_{180}\|/r_R$ of Table 1 of the paper at the fixed neighbour ranks $R = 1$, 10 and 100 for every dataset, since the paper's R differs across datasets. (H) A whitened distance in the head's input space: the Mahalanobis 10-NN rule with the covariance of the standardised bank features (10,000 images) and a ridge of 10^{-2} or 10^{-1} times the mean eigenvalue. Findings: a whitened distance sees the frame hardly better than the Euclidean one (flip AUROC 45.3–54.7% on the four natural-image datasets, and an inverted natural-OOD AUROC on ImageNet-64; 94.5–97.2% on CelebA and FashionMNIST), consistent with the linear read-out not being a matter of scaling the axes: the three discriminant directions are swamped by the thousands of other whitened directions that a distance weighs equally. At a fixed rank the displacement ratios keep the order of Table 1 apart from the near-tie between CIFAR-10 and SVHN (ImageNet-64 0.42 to FashionMNIST

2.06 at $R = 10$), so that order is a property of the features, not of the chosen R. The ID images peak on upright (77.7–94.5% on CIFAR-10, CIFAR-100 and ImageNet-64) and the upside-down ID images on a wrong angle (81.0–95.7%, mostly $180°$). The natural OOD images flagged by the orientation term only are read as confidently rotated (65.2–89.1%) rather than as frameless (flat posterior, 10.9–34.8%); of the images flagged by the kNN term only, the head reads 25.0–92.8% as upright and is undecided on most of the rest (flat, 5.3–61.4%). On CelebA and FashionMNIST almost no natural OOD image is flagged by the orientation term alone (4–8 of 4,500 and 2,000), and on SVHN 26–40, so their posterior statistics in Table S9 rest on few images.

	(H) whitened 10-NN				(F) shape of the posterior (% of images)							(G) $\|\Delta_{180}\|/r_R$		
	ridge 10^{-2}		ridge 10^{-1}		ID	upside-down ID		OOD, orient. only		OOD, kNN only				
ID dataset	flip	natural	flip	natural	flat	wrong	flat	flat	wrong	flat	wrong	$R = 1$	$R = 10$	$R = 100$
CIFAR-10	54.5	76.3	51.4	73.3	1.3	95.7	2.1	12.9	87.1	5.3	1.9	1.17	1.00	0.87
SVHN	54.7	99.7	53.9	99.7	0.7	46.8	3.2	17.4	82.6	34.6	22.1	1.17	1.03	0.91
CelebA	94.8	99.6	94.5	99.5	0.0	100.0	0.0	21.7	78.3	–	–	1.87	1.65	1.48
CIFAR-100	54.5	77.5	53.4	76.6	4.6	82.0	7.2	10.9	89.1	28.9	22.6	0.89	0.78	0.69
ImageNet-64	46.3	39.7	45.3	37.9	10.3	81.0	15.5	34.8	65.2	61.4	13.6	0.48	0.42	0.36
FMNIST	97.2	99.7	96.8	99.6	0.0	98.1	0.3	0.0	41.1	0.0	0.0	2.55	2.06	1.67

Table S9: Further controls of the two-notions analysis (mean over three noise seeds). (H): AUROC (%) of the Mahalanobis 10-NN rule in the standardised head space (flip: ID vs upside-down ID; natural: mean over the natural OOD sets). (F): share (%) of the images whose posterior over the four angles is flat (no angle above 0.5) or peaked on a wrong angle; the remainder is peaked on upright, which for images flagged by the orientation term occurs only where the ID threshold on $P(\mathrm{up} \mid \mathbf{x})$ exceeds 0.5 (CelebA, FashionMNIST). (G): median ratio of the displacement of a bank image under a half turn to its distance to its R-th nearest bank image, in the space of the kNN term (4,000 bank images). –: no image in that region.

Table S10: The two terms per pair (final evaluation images, three seeds): AUROC (%) of the kNN term, the orientation term and their sum (Upright kNN); fraction (%) of the OOD images flagged by the kNN term only, by the orientation term only and by both, at the thresholds that keep 95% of the ID images; Spearman correlation ρ of the two terms on the OOD images.

ID	OOD	kNN term	orient. term	sum	kNN only	orient. only	both	ρ
CIFAR-10	CIFAR-100	77.4	86.0	80.3	13.0	20.2	19.0	0.42
	SVHN	99.7	99.3	99.9	0.3	0.3	99.4	−0.17
	CelebA	89.9	97.2	96.0	2.9	62.7	22.6	0.10
	Textures	99.5	98.0	99.5	4.9	2.0	92.2	0.18
	Tiny-ImageNet	92.0	94.3	96.1	10.5	39.0	36.7	0.11
	LSUN-FIX	96.0	92.9	96.7	29.9	12.6	45.1	0.09
	ImageNet-FIX	96.9	94.5	97.6	19.8	10.3	61.2	0.25
	ImageNet32	96.0	90.4	96.3	26.2	8.8	50.5	0.36
	Places365	95.5	91.0	96.0	28.3	11.3	43.9	0.28
	upside-down ID	77.8	99.2	96.1	0.5	78.1	19.0	0.16
SVHN	CIFAR-10	99.1	96.7	99.6	15.0	1.4	82.3	0.44
	CIFAR-100	99.3	95.6	99.7	21.9	0.4	76.7	0.33
	CelebA	98.2	93.0	99.2	34.7	2.3	59.1	0.16
	Textures	99.8	93.8	99.9	31.2	0.0	68.5	0.18
	Tiny-ImageNet	99.3	95.0	99.7	24.8	0.9	73.8	0.42
	LSUN-FIX	99.7	97.1	99.9	15.7	0.5	83.5	0.47
	ImageNet-FIX	99.7	96.5	99.9	16.6	0.5	82.6	0.37
	ImageNet32	99.7	97.3	99.9	14.5	0.2	85.0	0.39
	Places365	99.7	96.7	99.8	16.4	0.3	82.9	0.52
	upside-down ID	56.1	78.6	64.5	4.7	25.1	2.6	0.12
CelebA	CIFAR-10	100.0	100.0	100.0	0.0	0.0	100.0	0.10
	CIFAR-100	100.0	100.0	100.0	0.0	0.0	100.0	0.13
	SVHN	100.0	100.0	100.0	0.0	0.0	100.0	0.15
	Textures	100.0	100.0	100.0	0.0	0.0	100.0	−0.08
	Tiny-ImageNet	99.9	100.0	100.0	0.0	0.4	99.6	0.10
	LSUN-FIX	99.9	100.0	100.0	0.0	0.3	99.7	0.14

ID	OOD	kNN term	orient. term	sum	kNN only	orient. only	both	ρ
	ImageNet-FIX	100.0	100.0	100.0	0.0	0.0	100.0	0.19
	ImageNet32	100.0	100.0	100.0	0.0	0.0	100.0	0.09
	Places365	100.0	100.0	100.0	0.0	0.2	99.8	0.24
	upside-down ID	99.5	100.0	100.0	0.0	0.0	100.0	0.23
CIFAR-100	CIFAR-10	66.3	28.4	64.1	12.0	0.6	0.7	0.29
	SVHN	98.8	95.1	99.5	34.7	4.1	59.8	−0.10
	CelebA	77.4	73.8	78.4	2.2	11.1	0.5	0.20
	Textures	98.7	90.7	99.0	55.5	2.4	36.7	0.03
	Tiny-ImageNet	82.2	82.3	86.4	13.5	27.0	5.1	−0.07
	LSUN-FIX	81.8	67.9	81.7	9.3	8.8	0.6	−0.06
	ImageNet-FIX	87.7	76.7	88.3	20.7	14.4	6.7	0.21
	ImageNet32	89.0	73.7	88.9	29.1	7.9	8.5	0.22
	Places365	83.2	67.1	83.0	13.5	9.3	4.1	0.17
	upside-down ID	56.5	95.0	79.1	1.6	69.4	3.5	−0.01
ImageNet-64	SSB-hard	67.8	58.7	67.3	14.3	4.5	0.9	0.48
	NINCO	68.3	72.1	71.7	8.9	13.7	0.6	0.06
	iNaturalist	68.1	75.1	71.9	1.9	14.8	0.7	0.05
	Textures	91.0	87.9	93.6	29.7	19.4	14.5	0.01
	OpenImage-O	70.1	73.0	73.2	7.6	12.8	1.3	0.12
	Places365	81.8	80.1	83.4	6.1	10.1	0.5	0.14
	CelebA	92.2	94.8	95.8	10.2	44.6	11.7	−0.06
	upside-down ID	50.5	95.5	80.4	3.8	69.3	1.5	−0.43
FMNIST	MNIST	100.0	100.0	100.0	0.0	0.0	100.0	0.30
	KMNIST	99.9	100.0	100.0	0.0	0.0	100.0	0.04
	Omniglot	99.8	100.0	99.9	0.0	1.5	98.5	0.38
	notMNIST	100.0	99.4	100.0	5.3	0.0	94.7	0.33
	upside-down ID	96.7	100.0	99.7	0.0	21.1	78.9	0.20

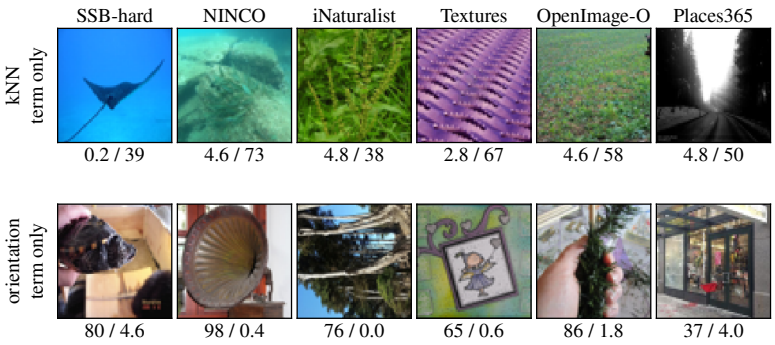


Figure S1: Natural OOD images of ImageNet-64 flagged by one notion only (final evaluation images, seed 0). Fixed rule: for each of the six non-face OOD sets, the image flagged by the kNN term alone that the orientation term finds most ID-like (top row), and the reverse (bottom row). Under each image: percentile of the kNN term / of the orientation term among the ID images; flagged: below 5. The kNN term alone flags images whose colour and texture statistics are unusual for ImageNet while the head still finds an upright frame; the orientation term alone flags images that the kNN term ranks as typical but in which the head finds no recognisable frame (close-ups, objects without a canonical pose, a view from below).

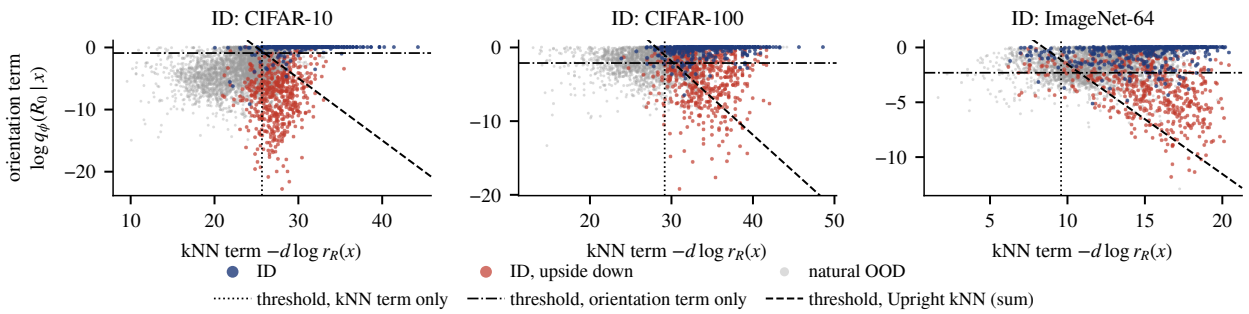


Figure S2: The two terms of Eq. (2) on the final evaluation images (noise seed 0) for the three ID datasets on which Upright kNN has the lowest AUROC; lines: thresholds keeping 95% of the ID images for the kNN term alone (dotted), the orientation term alone (dash-dotted; ceiling $\log 1 = 0$) and their sum (dashed). Upside-down ID images (red) overlap the ID images (navy) in the kNN term and fall below the dashed line through the orientation term; natural OOD images (grey) leave the ID region along both axes, many along one only.

S7 All Pairs

Table S11: AUROC (%) of every pair (final evaluation; Upright kNN, both kNN rules, CFS, GEPC and EigenScore averaged over three noise seeds). Best prior: the best of the prior methods of Table 2 on that pair (chosen after the fact, CSI included where it was run). Difference: Upright kNN minus that method; **bold** = win, *italic* = loss, plain = tie (bootstrap rule of the paper). By that rule Upright kNN loses to a prior detector run on all six ID datasets on SVHN→CIFAR-10, SVHN→CIFAR-100, SVHN→CelebA, ImageNet-64→iNaturalist; to CSI on CIFAR-10→CIFAR-100, FashionMNIST→Omniglot; and to its own tuned kNN rule on CIFAR-100→CIFAR-10. Three of the nine OOD sets of each 32×32 ID dataset (Tiny-ImageNet, ImageNet-FIX, ImageNet32) are resizings of ImageNet images, so the natural pairs are not independent.

ID	OOD	Ours	best prior		diff.	tuned kNN	diff.
CIFAR-10	CIFAR-100	80.3	91.1	CSI	*−10.9*	77.1	**+3.2**
	SVHN	99.9	99.9	CSI	0.0	99.7	+0.1
	CelebA	96.0	94.9	CSI	+1.1	89.6	**+6.4**
	Textures	99.5	99.3	CSI	+0.2	99.6	−0.1
	Tiny-ImageNet	96.1	93.6	CSI	**+2.5**	92.1	**+3.9**
	LSUN-FIX	96.7	93.5	CSI	**+3.2**	96.2	+0.6
	ImageNet-FIX	97.6	95.4	CSI	**+2.2**	97.1	+0.5
	ImageNet32	96.3	91.5	CSI	**+4.8**	96.2	+0.1
	Places365	96.0	92.7	CSI	**+3.4**	95.7	+0.3
	upside-down ID	96.1	99.3	Rot. pred.	*−3.2*	77.2	**+18.9**
	uniform noise	100.0	100.0	NLL	0.0	100.0	0.0
	constant	100.0	100.0	LID+NLL	0.0	100.0	0.0
SVHN	CIFAR-10	99.6	99.9	DDPM-OOD	*−0.3*	99.2	**+0.5**
	CIFAR-100	99.7	99.9	DDPM-OOD	*−0.2*	99.4	**+0.3**
	CelebA	99.2	100.0	NLL	*−0.8*	98.2	**+1.0**
	Textures	99.9	99.9	MSMA*	0.0	99.8	**+0.1**
	Tiny-ImageNet	99.7	99.8	DDPM-OOD	−0.1	99.4	**+0.4**
	LSUN-FIX	99.9	99.9	DDPM-OOD	0.0	99.7	**+0.2**
	ImageNet-FIX	99.9	99.9	DDPM-OOD	0.0	99.7	**+0.2**
	ImageNet32	99.9	100.0	DiffPath	−0.1	99.7	**+0.2**
	Places365	99.8	99.9	DDPM-OOD	−0.1	99.7	**+0.2**
	upside-down ID	64.5	84.6	CSI	*−20.0*	55.9	**+8.6**
	uniform noise	100.0	100.0	NLL	0.0	100.0	0.0
	constant	100.0	100.0	CSI	0.0	100.0	0.0
CelebA	CIFAR-10	100.0	100.0	CSI	0.0	100.0	0.0
	CIFAR-100	100.0	100.0	Rot. pred.	0.0	99.9	**+0.1**

ID	OOD	Ours	best prior		diff.	tuned kNN	diff.
	SVHN	100.0	100.0	DiffPath	0.0	100.0	0.0
	Textures	100.0	100.0	Rot. pred.	0.0	100.0	0.0
	Tiny-ImageNet	100.0	100.0	CSI	0.0	99.9	0.0
	LSUN-FIX	100.0	100.0	Rot. pred.	0.0	99.9	0.0
	ImageNet-FIX	100.0	100.0	Rot. pred.	0.0	100.0	0.0
	ImageNet32	100.0	100.0	Rot. pred.	0.0	99.9	+**0.1**
	Places365	100.0	100.0	Rot. pred.	0.0	100.0	0.0
	upside-down ID	100.0	100.0	Rot. pred.	0.0	99.8	+**0.2**
	uniform noise	100.0	100.0	NLL	0.0	100.0	0.0
	constant	100.0	100.0	IC-PNG	0.0	100.0	0.0
CIFAR-100	CIFAR-10	64.1	65.9	CFS-kNN	−1.8	66.6	*−2.5*
	SVHN	99.5	98.6	LMD	+**0.9**	98.9	+**0.6**
	CelebA	78.4	61.3	IC-J2K	+**17.1**	77.7	+0.7
	Textures	99.0	93.3	CFS	+**5.7**	98.8	+0.2
	Tiny-ImageNet	86.4	80.2	CSI	+**6.2**	82.6	+**3.8**
	LSUN-FIX	81.7	78.5	IC-PNG	+**3.1**	82.4	−0.7
	ImageNet-FIX	88.3	78.0	CSI	+**10.3**	88.1	+0.2
	ImageNet32	88.9	81.6	CFS-kNN	+**7.3**	89.4	−0.5
	Places365	83.0	79.0	IC-PNG	+**4.0**	83.7	−0.7
	upside-down ID	79.1	96.5	Rot. pred.	*−17.4*	56.8	+**22.4**
	uniform noise	100.0	100.0	NLL	0.0	100.0	0.0
	constant	100.0	100.0	MSMA*	0.0	100.0	0.0
ImageNet-64	SSB-hard	67.3	56.2	MSMA*	+**11.2**	67.8	−0.5
	NINCO	71.7	65.0	Rot. pred.	+**6.7**	68.3	+**3.4**
	iNaturalist	71.9	84.4	NLL	*−12.5*	68.1	+**3.8**
	Textures	93.6	92.0	CFS-kNN	+1.6	91.0	+**2.6**
	OpenImage-O	73.2	69.6	CFS-kNN	+**3.6**	70.1	+**3.1**
	Places365	83.4	74.8	MSMA*	+**8.7**	81.8	+**1.6**
	CelebA	95.8	76.5	MSMA*	+**19.3**	92.2	+**3.6**
	upside-down ID	80.4	94.5	Rot. pred.	*−14.1*	50.5	+**29.9**
FashionMNIST	MNIST	100.0	100.0	CFS-kNN	0.0	100.0	0.0
	KMNIST	100.0	100.0	CSI	0.0	99.9	+**0.1**
	Omniglot	99.9	100.0	CSI	*−0.1*	99.6	+**0.3**
	notMNIST	100.0	100.0	CFS-kNN	0.0	99.9	0.0
	upside-down ID	99.7	100.0	CSI	*−0.3*	96.0	+**3.7**
	uniform noise	100.0	100.0	NLL	0.0	100.0	0.0
	constant	100.0	100.0	IC-PNG	0.0	100.0	0.0

S8 FPR95

Method	CIFAR-10	SVHN	CelebA	CIFAR-100	ImageNet-64	FMNIST	Mean	Flip
Upright kNN	19.4	1.5	0.0	62.0	75.7	0.1	26.5	41.3
kNN (tuned)	28.8	1.7	0.0	63.3	84.5	0.4	29.8	64.2
kNN (basic)	54.3	8.9	0.0	69.0	88.6	1.3	37.0	71.9
NLL	86.7	3.4	41.0	91.3	88.9	98.8	68.3	90.4
IC-PNG	75.6	31.2	44.0	84.5	89.4	67.8	65.4	93.2
IC-J2K	72.3	52.5	53.4	85.3	93.5	95.8	75.5	92.6
TT	90.0	3.8	41.6	92.9	89.7	86.5	67.4	91.8
MSMA*	84.9	4.6	1.1	83.5	89.9	22.4	47.7	77.2
DiffPath	72.3	1.6	0.3	79.2	91.5	72.8	53.0	78.4
GEPC	90.3	51.1	4.2	90.0	92.9	77.2	67.6	95.8
EigenScore	85.5	61.0	55.6	94.3	95.6	27.5	69.9	74.4
DDPM-OOD	65.0	0.6	0.4	79.8	84.1	48.1	46.3	78.3
LMD	69.1	67.2	5.8	81.9	96.0	7.1	54.5	66.8
LID+NLL	–	–	–	–	–	–	–[§]	–
CFS	73.5	56.2	0.3	78.9	85.9	13.3	51.4	86.5
CFS-kNN	58.9	41.1	0.0	79.2	85.7	0.2	44.2	65.3
Rot. pred.	58.2	27.0	0.0	82.3	91.9	1.4	43.5	19.8
CSI	25.8	11.9	0.0	71.9	–	0.0	21.9[§]	23.4

Table S12: FPR at 95% TPR (%, lower is better; final evaluation), mean over the natural OOD sets of each ID dataset, their mean, and the mean over the upside-down sets. [§]Mean over the ID datasets on which the method was run; LID+NLL has no per-image score (its AUROC is a frontier over threshold pairs) and no FPR95.

S9 Ablations

Variant	CIFAR-10	SVHN	CelebA	CIFAR-100	ImageNet-64	FMNIST	Mean	Flip	Selection
Upright kNN, Eq. (2)	95.9	99.4	100.0	84.6	80.3	99.9	93.4	86.4	93.3
kNN term only	93.9	99.1	100.0	83.5	78.1	99.9	92.4	72.7	93.0
orientation term only	94.2	95.8	100.0	72.1	77.2	99.7	89.8	94.7	87.6
no temperature ($T = 1$)	95.9	99.3	100.0	84.6	80.3	99.9	93.3	87.7	93.2
Fisher (sum of log p-values)	96.5	99.4	100.0	82.3	80.6	100.0	93.1	92.3	92.5
rotated bank for $\bar{p}$ ($4R$, d re-estimated)	91.8	99.4	99.9	84.0	80.5	99.8	92.6	86.5	92.9
head trained with rotated noise*	94.5	99.5	100.0	83.6	80.1	99.9	92.9	78.8	93.1
orientation term over 4 test views*	95.3	99.1	100.0	84.3	79.0	99.9	92.9	88.1	93.2

Table S13: Development evaluation images, noise seed 0: mean AUROC over the natural OOD sets, the upside-down mean, and the mean on the selection data. No variant raises the selection-data mean, so within this table the frozen configuration would also be chosen on the selection data alone. The rotated-bank variant makes the kNN term closer to an estimate of $\bar{p}$ (the premise of Proposition 1) but is worse: on datasets where a rotation moves an image out of its neighbourhood (CIFAR-10), rotated bank images are foreign neighbours for upright test images, which blurs the kNN density more than it gains in rotation invariance. *Approximation from stored features whose noise was rotated with the image.

A second kNN term on the deepest decoder block (post hoc). Upright kNN uses one network evaluation. Allowing at most one more, we asked whether the depth that the mirror test ranks highest (Section S10; $1/4$ of the input resolution for the 32×32 datasets and $1/8$ for ImageNet-64, an 8×8 block in every case) carries information that the three blocks of the kNN term do not. The variant adds to Eq. (2) of the paper up to two further experts, each a kNN log-density on that block paired with the same orientation term,

$$s'(\mathbf{x}) = s(\mathbf{x}) + a_0\, e_{\sigma_1}(\mathbf{x}) + w^* e_{\sigma^*}(\mathbf{x}), \qquad e_\sigma(\mathbf{x}) = -d' \log r'_{R^*}(\mathbf{x};\sigma) + \log q_\phi(\mathbf{R}_0 \mid \mathbf{x}), \tag{2}$$

where $r'_{R^*}(\mathbf{x};\sigma)$ is the distance to the R^*-th nearest bank image in the pooled statistics of that block for the input $\mathbf{x} + \sigma\boldsymbol{\varepsilon}$, and d' the intrinsic dimension of that space. Every choice was made from ID images and the selection data

by rules written before the evaluation (`PROTOCOL_VL.md`; Table S16): the block by the mirror test; R^* by the rule that chose R for Upright kNN; $a_0 = 1$ if the expert at σ_1 raises the selection-data mean; σ^* the noise level among $\sigma_1, \ldots, \sigma_5$ (0.01 to 0.086) with the highest mirror test of the block, and $w^* = \max(0,\, 2M^* - 1)$, set to 0 when $\sigma^* = \sigma_1$ or when the bootstrap interval of M^* contains 50%; only $w^* > 0$ costs the second network evaluation (5 of the six datasets). Each expert contains a content density, so the variant is a product of experts [Hinton, 2002] that counts the content density more than once: it lies outside the decomposition of Proposition 1, which is why we report it as an ablation and not as the method. It was evaluated once on the final images after these rules were frozen, a second use of those images after the main evaluation.

Result (Table S14). The six-dataset mean rises from 93.4 to 94.6% (23/38/2 over the 63 pairs; mean FPR95 from 26.5 to 22.1%), on CIFAR-10, CIFAR-100 and ImageNet-64 only. On the near-OOD pairs, CIFAR-10→CIFAR-100 rises from 80.3 to 88.3% and CIFAR-100→CIFAR-10 from 64.1 to 68.8%; on ImageNet-64, SSB-hard and NINCO gain +2.0 and +1.7 points (Places365 +9.8). The two losses are CIFAR-100→CelebA (78.4 vs 76.4%), CIFAR-100→Textures (99.0 vs 98.8%). At 88.3% on CIFAR-10→CIFAR-100 the variant approaches CSI (91.1% in our run), which trains a ResNet-18 on the ID pixels, with a frozen denoiser and a small head.

Breakdown and two controls (development images, three noise seeds; Table S15). Raising the weight of the orientation term by the same amount instead changes nothing (93.3 against 93.3%): the gain is the kNN term of the deep block. The single-pass expert $a_0 e_{\sigma_1}$ alone gives 94.2%; the second evaluation adds little on average (94.3%) but lifts CIFAR-10→CIFAR-100 from 83.2 to 88.6%. Control 1 stays inside the decomposition: the densities of the deep features $\mathbf{a}$ and of the features of Upright kNN $\mathbf{b}$ are estimated as one density by the chain rule $p(\mathbf{a})\, p(\mathbf{b} \mid \mathbf{a})$ (a conditional kNN: the kNN of $\mathbf{b}$ among the K bank images nearest in $\mathbf{a}$, K chosen on the selection data), and the orientation term is added once. It reaches 94.1%, on par with the single-pass product of experts, with 6 losses against 2; the selection data chose the largest K (the weakest conditioning) for 5 of the six datasets, where the chain rule approaches adding the deep density separately. The gain is the information of the deep block, and the product-of-experts form is not the only way to add it. Control 2 adds a third expert on the channel means of the block: the hardest pair (CIFAR-100→CIFAR-10) rises from 65.3 to 71.1%, but the mean falls to 93.8% and the losses against Upright kNN rise to 17; more experts are not better.

ID dataset	Upright kNN	+ deep-block term	Δ [95% interval]	W/T/L	NFE
CIFAR-10	95.4	96.8	+1.4 [+1.1, +1.8]	7/5/0	2
SVHN	99.7	99.7	0.0 [0.0, 0.0]	1/11/0	2
CelebA	100.0	100.0	0.0 [0.0, 0.0]	0/12/0	2
CIFAR-100	85.5	87.4	+1.9 [+1.3, +2.5]	6/4/2	2
ImageNet-64	79.6	83.5	+4.0 [+3.2, +4.7]	8/0/0	1
FashionMNIST	100.0	100.0	0.0 [0.0, 0.0]	1/6/0	2
Mean	93.4	94.6	+1.2	23/38/2	–

Table S14: The deep-block ablation on the final images (AUROC, %, mean over the natural OOD sets of each ID dataset; three noise seeds; evaluated once after its rules were frozen, a second use of the final images). Δ: gain over Upright kNN with its bootstrap 95% interval; W/T/L: wins/ties/losses of the variant against Upright kNN over the pairs of the dataset by the bootstrap rule; NFE: network evaluations per image.

		near-OOD pairs				W/T/L		
	Mean	C10→C100	C100→C10	IN64→SSB	IN64→NINCO	vs Upright	vs variant	NFE
Upright kNN	93.3	81.0	60.8	69.8	71.4	–	–	1
+ deep-block term (frozen rule)	94.3	88.6	65.3	71.3	71.7	23/36/4	–	1–2
single pass only ($a_0 e_{\sigma_1}$)	94.2	83.2	64.8	71.3	71.7	20/41/2	5/47/11	1
orientation weight raised instead	93.3	82.2	59.2	68.8	72.2	13/40/10	3/39/21	1
Control 1: chain rule (conditional kNN)	94.1	82.5	65.7	71.7	71.1	16/41/6	6/43/14	1
Control 2: + third expert (channel means)	93.8	87.7	71.1	68.4	69.9	14/32/17	7/28/28	1–2

Table S15: Breakdown of the gain and two controls (development evaluation images, three noise seeds). Mean: six-dataset mean AUROC over the natural OOD sets; near-OOD pairs (C10: CIFAR-10, C100: CIFAR-100, IN64: ImageNet-64, SSB: SSB-hard); W/T/L over the 63 pairs against Upright kNN and against the variant with the frozen rule (bootstrap rule of the paper).

ID dataset	depth	M_0	R^*	a_0	σ^*	M^* [95% interval]	w^*	NFE
CIFAR-10	1/4	87.7	2	1	$\sigma_5 = 0.086$	92.9 [91.8, 94.0]	0.86	2
SVHN	1/4	62.0	5	0	$\sigma_4 = 0.050$	64.0 [61.7, 66.5]	0.28	2
CelebA	1/4	99.8	5	1	$\sigma_3 = 0.029$	100.0 [99.9, 100.0]	1.00	2
CIFAR-100	1/4	63.3	2	1	$\sigma_2 = 0.017$	61.6 [59.2, 64.0]	0.23	2
ImageNet-64	1/8	62.1	1500	1	$\sigma_5 = 0.086$	50.7 [48.3, 53.3]	0.00	1
FashionMNIST	1/4	98.2	5	0	$\sigma_5 = 0.086$	98.3 [97.9, 98.6]	0.97	2

Table S16: The frozen choices of the deep-block ablation (noise seed 0, as recorded in `PROTOCOL_VL.json`; the mirror tests here were computed in half precision, those of Table S17 in single precision). Depth: resolution ratio of the block; M_0: its mirror test (σ_1, $R = 1$); M^*: the mirror test at σ^* and R^* with its 95% interval; NFE: network evaluations per image.

S10 Mirror Test

	vertical M by depth				horizontal M (control)				kNN term of ours			
ID	1	1/2	1/4	1/8	1	1/2	1/4	1/8	vert.	horiz.	ρ	ρ_{H}
CIFAR-10	52.1	67.3	87.8	–	50.0	52.8	54.2	–	77.7	52.4	0.91(29)	0.61
SVHN	50.4	51.4	61.9	65.3	50.2	52.0	64.1	69.1	55.7	57.2	−0.53(13)	−0.56
CelebA	58.8	97.9	99.8	97.1	49.6	50.3	51.0	51.1	99.4	50.3	0.91(33)	0.46
CIFAR-100	51.4	56.3	63.3	–	50.2	52.2	55.7	–	56.6	52.6	0.68(29)	0.68
ImageNet-64	49.9	51.1	53.8	62.1	50.0	50.1	49.4	49.9	50.2	49.9	0.89(30)	−0.14
FashionMNIST	69.4	93.7	98.2	95.8	59.1	72.4	80.6	77.4	96.6	77.0	0.87(22)	0.84

Table S17: Mirror test M: AUROC (%) of ID development images against their vertical flips under the kNN term ($R = 1$) of the last decoder block at each depth (resolution ratio; –: no such level; seed 0, σ_1), with the horizontal flip as control (Proposition 2 says the horizontal flip should be invisible for left–right symmetric data); the kNN term of Upright kNN (three blocks, frozen R); and the Spearman correlation over all blocks (number in parentheses) between the vertical M and the natural-OOD AUROC of the block (ρ), and the same for the horizontal M (ρ_{H}).

S11 ImageNet-1K at 64×64 by Set

Method	SSB-hard	NINCO	iNat	Textures	OI-O	V2	C	R	FPR near	FPR far	W/T/L
Upright kNN	68.6	71.9	70.1	93.2	73.9	52.4	92.3	77.9	83.9	74.4	–
kNN term	69.7	69.3	66.2	91.4	71.5	52.2	92.1	73.4	82.2	76.6	4/0/1
orient. term	56.4	70.6	74.4	86.1	71.9	50.9	80.7	77.8	91.4	81.0	2/2/1
kNN (basic)	56.0	65.0	46.8	89.9	67.5	50.6	83.6	63.9	87.8	74.2	5/0/0
CFS (diag.)	57.5	63.8	68.5	88.7	67.6	52.0	77.3	68.7	86.0	76.0	4/1/0
CFS (kNN)	52.3	60.4	53.4	90.3	69.4	51.5	80.7	67.0	92.4	71.2	5/0/0
Rot. pred.	43.8	62.4	67.7	78.5	64.9	49.7	63.2	65.1	93.2	89.4	4/1/0
NLL	49.1	51.8	83.8	52.9	58.2	51.0	43.3	31.3	93.2	79.0	4/0/1
IC-PNG	49.8	55.7	65.5	47.8	57.9	48.1	56.6	45.7	92.4	87.5	5/0/0
IC-J2K	49.0	53.4	65.0	46.5	55.6	48.3	59.9	36.1	93.5	91.8	5/0/0
TT	53.6	59.8	74.0	54.7	61.5	49.6	66.1	62.7	90.2	89.9	4/0/1
MSMA*	56.8	56.7	41.2	81.5	56.7	46.9	79.1	72.1	90.3	82.3	5/0/0
DiffPath	50.4	56.3	59.4	70.4	58.0	52.1	57.6	69.6	91.6	87.5	5/0/0
GEPC	46.3	53.7	46.2	65.4	57.7	49.8	57.8	59.0	94.1	87.2	5/0/0
EigenScore	47.1	45.6	42.3	48.9	46.9	49.5	54.0	67.8	96.8	95.1	5/0/0

Table S18: OpenOOD v1.5 ImageNet-1K, 64×64, EDM2-XS (AUROC and mean FPR95, %; 1,000 common images per set; all detectors, including those left out of Table 3 of the paper). W/T/L of Upright kNN over the five OOD sets.

Method	1	2	3	4	5
Upright kNN	88.4	91.3	91.6	93.3	94.5
kNN term	88.9	91.1	91.5	92.9	93.7
orient. term	74.9	78.6	79.5	82.0	84.0
kNN (basic)	75.3	81.9	82.5	86.8	88.3
CFS (diag.)	65.2	72.1	76.2	81.3	83.9

Table S19: AUROC (%) of ImageNet val against ImageNet-C by corruption severity (64×64, full sets, three seeds). Density-based scores flag corrupted images more as the severity grows.

Bank size (post hoc, `PROTOCOL_E5B.md`). The frozen bank of ImageNet-64 holds 10 training images per class. Table S20 repeats the benchmark with banks of 25,000 and 50,000 training images drawn at random from the training set with the same preprocessing as the benchmark images, with the head retrained on the new bank, R scaled with the bank ($R = N/10$, which reproduces the frozen $R = 1{,}000$ at $N = 10{,}000$) or $R = 1{,}000$ kept fixed, and CFS fitted on the same bank (three noise seeds; the protocol was written before the result). A larger bank changes neither term: 70.6/79.5% and 70.8/79.4% near/far AUROC with 25,000 and 50,000 images against 70.6/79.3% with the frozen bank (kNN term 70.1/76.6% and 70.1/76.7% against 70.0/76.7%), and keeping $R = 1{,}000$ with the larger banks costs up to two near-OOD points (70.0% and 68.8%). The near-OOD AUROC of ImageNet-64 is therefore not limited by the size of the bank.

	AUROC		FPR95		csID			Full spectrum	
Method	Near	Far	Near	Far	V2	C	R	Near	Far
bank of 10,000 (10 per class; the paper) training images									
Upright kNN ($R = N/10$)	70.6	79.3	83.8	75.5	51.7	91.8	76.5	55.2	64.3
$R = 1000$	70.6	79.3	83.8	75.5	51.7	91.8	76.5	55.2	64.3
kNN term ($R = N/10$)	70.0	76.7	84.5	80.1	51.6	91.6	71.9	56.3	63.0
orient. term	64.2	76.9	91.7	82.3	51.2	79.8	77.4	51.0	64.7
kNN (basic)	61.3	67.9	89.5	77.3	50.6	82.9	65.0	52.4	59.9
CFS (diag.)	59.9	74.1	85.5	76.4	50.7	75.7	66.5	51.6	65.2
CFS (kNN)	56.6	71.0	93.8	74.0	50.3	79.2	67.3	47.2	63.0
bank of 25,000 (random) training images									
Upright kNN ($R = N/10$)	70.6	79.5	83.6	75.7	51.7	91.8	75.9	55.4	64.6
$R = 1000$	70.0	79.3	86.8	74.2	51.9	89.0	75.6	55.0	65.6
kNN term ($R = N/10$)	70.1	76.6	84.4	80.2	51.7	91.5	72.0	56.3	62.9
orient. term	63.9	78.4	91.1	80.7	51.3	80.5	78.3	50.3	65.9
kNN (basic)	60.6	67.4	89.6	76.6	50.7	81.1	63.8	52.4	60.1
CFS (diag.)	59.9	74.1	85.4	76.3	50.7	75.7	66.6	51.6	65.2
CFS (kNN)	55.7	70.3	94.2	74.1	50.3	78.9	67.4	46.3	62.3
bank of 50,000 (random) training images									
Upright kNN ($R = N/10$)	70.8	79.4	83.7	76.1	51.7	91.8	75.4	55.7	64.8
$R = 1000$	68.8	79.3	88.3	72.4	52.1	87.0	74.4	54.7	66.9
kNN term ($R = N/10$)	70.1	76.7	84.4	80.1	51.7	91.5	72.0	56.4	63.0
orient. term	63.8	78.9	90.4	80.6	51.1	80.2	77.8	50.6	66.9
kNN (basic)	60.0	67.4	89.5	76.2	50.8	80.0	62.9	52.2	60.4
CFS (diag.)	59.9	74.1	85.4	76.3	50.7	75.7	66.5	51.6	65.2
CFS (kNN)	54.9	70.0	94.6	74.1	50.3	78.8	67.0	45.7	62.1

Table S20: OpenOOD v1.5 ImageNet-1K at 64×64 (full sets, 42,000 ID images; AUROC and FPR95, %; three noise seeds) as the bank grows (`PROTOCOL_E5B.md`).

S12 ImageNet at Higher Resolution with Resolution Control

Every image (ID and OOD) is first resized so that its short side is 224 pixels and centre-cropped to 224×224, then resized to the model's resolution, so that no set can be told apart by the resolution at which it is stored (protocols

PROTOCOL_HIRES.md and PROTOCOL_HIRES_DIT.md, written before the runs). The ImageNet-64 configuration of Upright kNN is used unchanged ($R = 1{,}000$ of a bank of 10 training images per class, σ_1, three blocks, the head and its training); for the latent models the noise is added to the latents and the bank images are rotated in pixel space before encoding. CFS is not defined for a transformer; MSMA*, DiffPath and GEPC are run in pixel space only (ADM).

	ADM 256			EDM2-XS 512			DiT-XL/2 256		
Method	Near	Far	Flip	Near	Far	Flip	Near	Far	Flip
Upright kNN	**83.7**	**76.4**	**71.2**	**64.7**	**74.5**	**77.3**	**58.3**	**71.8**	**77.4**
kNN term	84.1	75.3	50.1	63.6	71.3	49.9	56.5	67.7	51.5
orient. term	65.4	69.3	91.3	62.2	75.4	94.2	61.5	78.6	96.9
kNN (basic)	78.6	53.9	50.0	57.5	68.8	50.0	57.5	67.6	52.0
CFS (diag.)	66.3	**76.4**	49.2	61.2	72.7	49.2	–	–	–
CFS (kNN)	66.7	62.7	50.6	56.4	66.0	50.9	–	–	–
MSMA*	69.4	67.6	50.4	–	–	–	–	–	–
DiffPath	54.0	48.0	49.9	–	–	–	–	–	–
GEPC	46.6	51.3	50.1	–	–	–	–	–	–

Table S21: ImageNet-1K at higher resolution with three further diffusion models (ADM in pixels; EDM2-XS and DiT-XL/2 in latent space), every image passed through the same 224-pixel bottleneck (AUROC, %; frozen ImageNet-64 configuration; three noise seeds). Near / far as in Table 3 of the paper; Flip: upside-down ID images. –: method not run for that model. Best and second best per column (single-term rows excluded) are marked as in the paper.

Method	SSB-hard	NINCO	iNat	Tex.	OI-O	Flip	V2	C	R	W/T/L
ADM, 256×256, pixels: 500–500 common images per set										
Upright kNN	87.0	80.3	67.4	86.4	75.4	71.2	55.3	94.3	84.8	–
kNN term	88.4	79.9	67.7	85.4	72.8	50.1	54.5	94.8	83.6	2/2/1
orient. term	64.4	66.4	56.9	76.3	74.6	91.3	52.8	73.4	73.4	4/1/0
kNN (basic)	83.2	74.0	77.2	61.2	23.3	50.0	56.1	84.6	68.1	4/0/1
CFS (diag.)	67.7	64.8	59.2	90.0	79.9	49.2	45.8	94.3	68.2	3/0/2
CFS (kNN)	68.2	65.3	51.8	85.0	51.3	50.6	49.3	83.6	70.3	4/1/0
MSMA*	73.9	64.9	35.1	87.1	80.5	50.4	46.8	92.5	79.3	3/1/1
DiffPath	55.9	52.0	33.3	59.4	51.4	49.9	46.2	78.4	65.4	5/0/0
GEPC	44.1	49.1	62.1	48.5	43.3	50.1	47.0	64.4	46.1	5/0/0
EDM2-XS, 512×512, latents: 1000–1000 common images per set										
Upright kNN	62.0	67.3	64.9	87.3	71.4	77.3	50.3	87.6	78.1	–
kNN term	62.9	64.3	60.9	84.2	68.8	49.9	50.3	86.9	74.6	4/0/1
orient. term	55.0	69.4	72.1	82.9	71.1	94.2	50.3	77.7	77.2	2/1/2
kNN (basic)	56.4	58.6	51.5	85.7	69.2	50.0	51.6	87.1	59.7	5/0/0
CFS (diag.)	58.8	63.6	64.3	88.6	65.2	49.2	50.1	84.5	69.7	3/1/1
CFS (kNN)	54.5	58.3	43.6	87.4	67.1	50.9	51.6	86.9	72.7	4/1/0
DiT-XL/2, 256×256, latents: 1000–1000 common images per set										
Upright kNN	53.6	63.1	60.9	87.2	67.3	77.4	48.7	76.1	75.0	–
kNN term	54.1	58.8	54.9	84.7	63.6	51.5	48.7	73.6	71.1	4/1/0
orient. term	51.1	71.8	75.6	86.6	73.6	96.9	49.9	76.6	78.9	1/1/3
kNN (basic)	55.9	59.1	51.7	90.0	61.2	52.0	50.1	81.3	66.1	3/0/2

Table S22: Resolution-controlled ImageNet by set (AUROC, %; common images of all rows of a model; three noise seeds for Upright kNN, the kNN rules and CFS). W/T/L of Upright kNN over the five OOD sets (bootstrap rule of the paper).

Diagnostics (post hoc). Simple pixel statistics of the bottlenecked 224-pixel images, the mean absolute Laplacian (high-frequency energy) and the excess absolute difference across 8-pixel column boundaries (JPEG blockiness), separate ImageNet from SSB-hard with at most 56.7% AUROC and from any OOD set with at most 62.9% (either direction). In the space of the kNN term of DiT-XL/2, the largest channel of each block carries 70–94% of the squared

norm of the pooled statistics (median over bank images; the same channel for nearly every image), and two random bank images have a median cosine similarity of 0.995 (0.84–0.87 for ADM and EDM2-XS); a quarter of the bank points therefore have degenerate neighbourhoods in the estimate of d, which skips them. **Identical final resampling (added after the frozen evaluation, expectation written before the run; `PROTOCOL_HIRES2.md`).** The 224-pixel bottleneck does not equalise what happened to an image before it: ImageNet validation images (median short side 375 pixels) are downsampled by our pipeline, whereas SSB-hard and ImageNet-C are stored at 224 pixels and are not resampled at all, and a denoiser at its finest noise level is sensitive to such spectral signatures. As a control written before its result, every 224-pixel image (bank, ID, OOD and csID) is reduced to 112 pixels by one $2\times$ box reduction and brought back to 256 pixels by one bicubic resize, so that every set has the same final resampling and no content above the 112-pixel Nyquist limit; the frozen ImageNet-64 configuration of Upright kNN, kNN (basic) and CFS are then run with ADM as before (three noise seeds; the export step of Upright kNN initially failed on the new tier name and was re-run unchanged). Table S23 compares the two tiers on the same 1,000 images per set with a paired bootstrap. The expectation written in advance was that a resampling signature would bring the near-OOD AUROC of Upright kNN down to its 64×64 values (SSB-hard 68.6%, NINCO 71.9%) and remove the inverted kNN (basic) result on OpenImage-O. SSB-hard stays at 87.7% (tie), so its gain at 256 pixels survives this control and is unlikely to be a resampling signature; NINCO falls from 80.5% to 69.0%, close to its 64×64 level (71.9%), and iNaturalist from 67.4% to 51.9%, chance, whereas OpenImage-O rises from 75.4% to 85.1% and the kNN (basic) inversion disappears (24.2% to 65.0% on these 1,000 images). CFS also loses on NINCO and keeps its SSB-hard AUROC (near-OOD mean 68.0% to 67.1%), and under the control Upright kNN still leads it on near-OOD (78.4% against 67.1%; W/T/L 3/0/2 over the five sets, the losses on iNaturalist and Textures). Part of the 256-pixel results above therefore depends on how the sets were stored and resampled, in both directions, and the SSB-hard result does not. The control removes content above 112 pixels together with the resampling history, and the 112-pixel images are blurred for a model trained on sharp images; it cannot separate these two effects, so the 256-pixel numbers of the paper should be read with this caveat.

	SSB-hard		NINCO		Near		iNat		Tex.		OI-O		Far		Flip	
Method	224	112	224	112	224	112	224	112	224	112	224	112	224	112	224	112
Upright kNN	87.7	87.7	80.5	69.0^{-}	84.1	78.4^{-}	67.4	51.9^{-}	86.8	86.9	75.4	85.1^{+}	76.5	74.6^{-}	70.8	64.6^{-}
kNN term	89.0	88.5	80.1	68.8^{-}	84.5	78.6^{-}	67.7	50.5^{-}	85.8	86.3	72.6	85.4^{+}	75.4	74.0^{-}	50.0	49.8
orient. term	65.6	63.4^{-}	66.7	63.1^{-}	66.2	63.3^{-}	58.3	63.0^{+}	77.0	75.2^{-}	75.0	68.7^{-}	70.1	69.0	90.9	87.8^{-}
kNN (basic)	84.0	82.4	75.7	65.3^{-}	79.8	73.9^{-}	78.8	47.2^{-}	62.6	77.9^{+}	24.2	65.0^{+}	55.2	63.4^{+}	50.2	49.9
CFS (diag.)	69.9	70.5	66.1	63.6^{-}	68.0	67.1^{-}	60.5	61.7^{+}	89.8	88.7^{-}	80.6	70.5^{-}	77.0	73.7^{-}	49.3	49.4
CFS (kNN)	69.4	66.0^{-}	67.7	60.7^{-}	68.5	63.3^{-}	54.2	40.8^{-}	85.6	84.2^{-}	54.1	65.1^{+}	64.6	63.4^{-}	50.5	50.4

Table S23: ADM 256×256 with the resolution-controlled images of Section S12 (224: every image through the 224-pixel bottleneck) and with the same final resampling for every set (112: every image through 112 pixels; `PROTOCOL_HIRES2.md`), on the same 1,000 images per set (AUROC, %; three noise seeds). $^{+}$/$^{-}$: the 112-pixel value is higher / lower than the 224-pixel one by the paired bootstrap rule of the paper.

S13 ImageNet at 256×256 without Resolution Control

Method	SSB-hard	NINCO	iNat	Textures	OI-O	Flip
Upright kNN	93.7	80.6	84.1	78.5	54.0	78.5
kNN term	94.9	81.4	86.2	75.1	48.6	50.6
orient. term	75.3	66.1	61.3	77.2	71.5	92.6
kNN (basic)	78.5	66.1	85.6	50.3	13.0	50.3
CFS (diag.)	85.4	62.5	57.3	88.5	76.5	49.4
CFS (kNN)	82.8	65.9	57.9	82.1	41.6	50.7
MSMA*	63.7	62.0	33.9	88.2	79.5	50.0
DiffPath	62.6	53.7	39.3	60.6	44.9	50.8
GEPC	70.6	54.0	48.4	49.1	40.0	50.8

Table S24: ADM 256×256 with each set's images cropped from their stored resolution (500 common images per set). SSB-hard is stored at 224 pixels (short side; checked on a sample of 200 images) and is upsampled, whereas ImageNet validation has a median short side of 375 pixels (3% of its images are below 256); ImageNet-C is also stored at 224 pixels. Every feature-based detector separates SSB-hard far better than at 64×64, a resolution shortcut; the paper therefore reports the resolution-controlled setting of Section S12, in which every image passes through the same 224-pixel bottleneck.

S14 Fusions

Score	Mean	Flip	W/T/L of ours
Upright kNN (sum, Eq. 2)	**93.4**	86.6	–
kNN term only	92.5	72.8	28/18/1
orientation term only	89.9	94.7	29/14/4
rotation classifier (ResNet-18)	84.0	94.6	35/12/0
kNN + rot. classifier, z-sum	91.2	92.4	30/17/0
kNN + rot. classifier, Fisher	91.2	92.9	19/26/2
kNN + orient. term, z-sum	<u>93.1</u>	92.4	23/17/7
kNN + orient. term, Fisher	<u>93.1</u>	92.5	13/26/8
Upright kNN with $d = 1$	92.0	94.2	24/20/3

Table S25: Principled sum vs. naive fusions (final evaluation, AUROC, %; Mean and Flip as in Table 2 of the paper). z-sum: sum of z-scores; Fisher: sum of log empirical ID p-values; both normalise each score with additional ID images (the development split), which Upright kNN does not need. W/T/L over the natural-OOD pairs. This comparison was added after the frozen evaluation and reuses its images.

The frame term as a plug-in (post hoc, `PROTOCOL_E9.md`, expectations written before the result). If the frame is a second notion of ID conformity, the orientation term should improve other neighbourhood detectors too, with the same parameter-free weight: the kNN log-density $-d \log r$ of that detector's own space, with d the Levina–Bickel estimate in that space (20 neighbours of 3,000 bank points), plus $\log q_\phi(\mathbf{R}_0 \mid \mathbf{x})$ of the paper's frozen head. Table S26 applies this to CFS with its kNN head (r: its mean distance to the 10 nearest bank descriptors in its standardised space, $d = 17$–37) and to the basic kNN of the paper (r: the chordal distance of the 10th neighbour in its half-resolution block, $d = 14$–24), and, as a control for the scale argument of Section 3.3, to the diagonal head of CFS read as a Gaussian log-density of all its descriptor dimensions (which ranks as the official slot-mean score except on SVHN, whose two slots have 1,024 and 256 dimensions). It also reports a z-sum (scales fitted on the ID images of the development split, as above) of the orientation term with every comparison method of Table 2 that was run on those images. Findings: with the parameter-free weight the frame term lifts CFS (kNN) from 85.5% to 88.6% (W/T/L 41/5/1 over the 47 natural pairs; 32 wins of at least one point) and the basic kNN from 87.7% to 89.8% (34/12/1), more than the 0.9 points it adds to the paper's stronger kNN term, and it leaves the Gaussian log-density of CFS (diag.) unchanged (80.5% to 80.7%: its values spread over hundreds of nats across ID images, the frame factor over one). The fitted z-sum adds +3.6 to +25.0 points to every one of the 12 comparison methods, but a fusion exceeds the orientation term alone (89.9%) only

for the two neighbourhood detectors (CFS (kNN), kNN (basic): 91.3% and 90.6%); for the likelihood and statistic detectors the fused score is carried by the frame term (NLL 62.7% to 87.8%, below 89.9%). The frame term is thus complementary to neighbourhood conformity in particular, as Section 5 argues, and not only to our features.

Score	CIFAR-10	SVHN	CelebA	CIFAR-100	ImageNet-64	FMNIST	Mean	Flip	W/T/L vs base
kNN term of Upright kNN	93.7	99.4	100.0	85.0	77.0	99.9	92.5	72.8	–
+ frame term (Bayes weight; Upright kNN)	95.4	99.7	100.0	85.5	79.6	100.0	93.4	86.6	28/18/1 (11/1)
kNN (basic)	88.6	98.6	100.0	80.3	59.1	99.7	87.7	68.2	–
+ frame term (Bayes weight)	92.3	99.5	100.0	81.6	65.6	99.9	89.8	84.7	34/12/1 (19/1)
+ frame term (z-sum)	93.6	99.3	100.0	81.3	73.8	99.9	91.3	91.7	22/23/2 (15/2)
CFS (kNN)	84.9	92.1	99.9	72.3	63.9	99.9	85.5	70.6	–
+ frame term (Bayes weight)	90.8	97.0	100.0	74.9	69.3	100.0	88.6	85.9	41/5/1 (32/1)
+ frame term (z-sum)	93.2	97.6	100.0	77.1	75.5	100.0	90.6	92.7	39/7/1 (31/1)
CFS (diag.)	73.8	90.7	99.7	65.3	67.2	97.6	82.4	59.3	–
+ frame term (z-sum)	89.5	97.5	100.0	73.1	76.6	99.9	89.4	91.9	42/4/1 (33/1)
CFS (diag.) as a Gaussian log-density	73.8	79.6	99.7	65.3	67.2	97.6	80.5	59.2	–
+ frame term (unit weight)	74.1	80.1	99.7	65.4	67.3	97.7	80.7	60.5	33/14/0 (0/0)
NLL	58.2	98.9	79.9	53.8	52.3	33.3	62.7	56.6	–
+ frame term (z-sum)	88.3	99.7	100.0	70.6	71.7	96.4	87.8	91.0	42/4/1 (38/1)
IC (PNG)	73.4	92.1	87.3	66.5	47.1	86.2	75.4	57.1	–
+ frame term (z-sum)	91.1	98.2	100.0	75.6	68.3	99.6	88.8	91.8	44/2/1 (43/1)
IC (JPEG2K)	71.4	74.1	82.0	65.2	46.4	40.8	63.3	56.6	–
+ frame term (z-sum)	90.1	94.9	100.0	74.2	68.2	96.6	87.3	91.8	45/1/1 (44/1)
TT	56.8	98.6	85.1	51.4	58.1	61.6	68.6	53.9	–
+ frame term (z-sum)	86.5	99.7	100.0	67.4	74.2	97.3	87.5	92.0	45/1/1 (43/1)
MSMA*	63.6	99.1	99.7	61.3	63.2	96.4	80.5	63.3	–
+ frame term (z-sum)	87.3	99.7	100.0	71.0	74.3	99.7	88.6	92.2	42/4/1 (30/1)
DiffPath	71.0	99.5	99.9	64.9	54.9	80.2	78.4	62.3	–
+ frame term (z-sum)	89.1	99.8	100.0	72.6	73.5	98.9	89.0	92.8	38/8/1 (26/1)
GEPC	51.9	91.3	99.1	55.6	51.2	84.4	72.3	52.1	–
+ frame term (z-sum)	84.4	97.7	100.0	69.5	71.8	98.7	87.0	92.0	45/1/1 (41/1)
EigenScore	73.4	85.8	86.9	47.6	44.2	94.3	72.0	75.4	–
+ frame term (z-sum)	91.6	97.3	100.0	65.2	65.6	99.6	86.5	93.9	46/0/1 (46/1)
Rotation pred.	85.3	95.4	100.0	61.8	62.1	99.4	84.0	94.6	–
+ frame term (z-sum)	92.0	97.3	100.0	71.0	73.1	99.8	88.9	96.0	36/10/1 (33/1)

Table S26: The frame term added to other detectors (final evaluation, AUROC, %, mean over the natural OOD sets of each ID dataset; Flip: upside-down ID images; three noise seeds for Upright kNN, CFS, the basic kNN and the frame term, one for the other rows). Bayes weight: $-d \log r + \log q_\phi(\mathbf{R}_0 \mid \mathbf{x})$ with d estimated in the detector's own space; unit weight: the Gaussian log-density plus $\log q_\phi$; z-sum: scales from the ID images of the development split. W/T/L: the row against its base over the 47 natural pairs (bootstrap rule of the paper), in parentheses wins/losses of at least one point.

S15 CSI

CSI [Tack *et al.*, 2020] is run with the official code (commit 60742b6, unmodified) in the unlabeled multi-class setting of its repository: ResNet-18, simclr_CSI with rotations as shifting transformations, 1,000 epochs, LARS with learning rate 0.1, 10 warm-up epochs, cosine schedule, weight decay 10^{-6}, temperature 0.5, on one GPU with a batch of 128 images (the official total batch of 512 rotated images). The training images are the bank images of each 32×32 ID dataset, i.e. the images of the kNN bank and of the rotation head of Upright kNN. The score is the official CSI score with the repository's evaluation flags (10 augmented samples, crop scale 0.54). A thin wrapper feeds our arrays and replaces two packages that do not build with our PyTorch version (a re-implementation of the LARS optimiser of torchlars 0.1.2 and a stub of the multi-GPU gather, which a single-GPU run does not call). The protocol was written before training. Training took 36–44 hours per ID dataset on one RTX 2080 Ti; scoring the 40 test views of each image and the training set took 5–18 minutes per ID dataset. On our 500-image final split the CIFAR-10 run gives 91.1% on CIFAR-100, 93.5% on LSUN-FIX and 95.4% on ImageNet-FIX, against the published 89.2, 90.3 and 93.3 (Table 2(a) of Tack *et al.* [2020]; five trials, standard deviations ≤ 0.3), i.e. two to three points above on each, which we attribute to the evaluation images. The CIFAR-100 run trained normally (final contrastive loss 5.26 against 5.20 for CIFAR-10; final shifting-transformation loss 0.16 against 0.09, consistent with the lower rotation accuracy of our head

on CIFAR-100) and its score detects flipped CIFAR-100 images at 89.1% AUROC, yet it ranks CelebA faces (9.8%) and uniform noise (58.5%) as more typical than CIFAR-100 test images; CSI's paper does not report CIFAR-100 as ID, so we have no published value to compare with. Table S11 lists CSI wherever it is the best prior method of a pair; its per-ID means are in Table 2 of the paper.

S16 EuroSAT, Chest X-Rays, Histology and Generators Trained with Rotations

Models. The EuroSAT [Helber *et al.*, 2019], ChestMNIST and PathMNIST [Yang *et al.*, 2023] models and the CIFAR-10 generators of Table S30 are EDM-style U-Nets (39M parameters) trained by us with the recipe of the FashionMNIST model (Section S2; 8M images seen for the three domains, 4M for the CIFAR-10 generators, with 5,000 images held out for calibration). All protocols (`PROTOCOL_E3.md`, `PROTOCOL_E4.md`) were written before any result, with $R = 5$ fixed in advance and no selection on these data; Upright kNN is otherwise unchanged. The chest X-ray tier uses 1,000 test radiographs as ID with their 90° and 180° rotations, vertical flips and left–right mirrors (a mirror is not in the rotation group and is reported only), other modalities (abdominal CT, greyscale histology), paediatric chest X-rays from another source (reported separately) and four natural sets in greyscale, 500 images each. The histology tier uses 1,000 test tiles with the same four transforms, other modalities (blood-cell microscopy, dermatoscopy, chest X-rays in RGB) and the ten natural sets of the CIFAR-10 tier.

Method	C10	C100	SVHN	CelebA	Tex.	TIN	LSUN-FIX	IN-FIX	IN32	P365	Mean	Flip	Rot. 90
Upright kNN	100.0	100.0	99.3	100.0	99.5	99.9	100.0	99.9	100.0	100.0	99.8	64.0	75.8
kNN term	99.8	99.8	99.7	99.9	99.6	99.8	99.8	99.8	99.8	99.8	99.8	54.2	55.1
orient. term	99.3	98.9	81.8	99.7	91.4	97.3	99.2	96.9	98.6	98.1	96.1	75.6	91.1
kNN (basic)	99.6	99.7	99.8	99.8	99.6	99.6	99.6	99.6	99.7	99.6	99.7	52.8	54.0

Table S27: EuroSAT as ID (three seeds): AUROC (%) against ten natural OOD sets (500 images each), and of ID images against their vertical flips and 90° rotations. Held-out rotation accuracy of the head 72–74%, temperature 0.91–0.94.

	ID transformed				other modality		other source	natural	
Method	rot. 90	rot. 180	upside down	mirror	CT	histology	paediatric CXR	mean (4)	W/T/L
Upright kNN	100.0	100.0	100.0	72.6	100.0	100.0	99.5	100.0	–
kNN term	98.6	99.8	99.8	71.7	100.0	100.0	99.6	100.0	4/7/0
orient. term	100.0	100.0	100.0	71.4	100.0	100.0	89.9	99.9	1/10/0
kNN (basic)	99.7	99.9	99.8	68.3	99.9	100.0	99.4	99.9	5/6/0
CFS (diag.)	98.6	96.5	96.4	49.6	99.4	99.8	91.8	99.9	9/1/1
CFS (kNN)	99.5	99.8	99.8	67.0	99.9	100.0	99.4	100.0	4/7/0
NLL	88.9	88.7	88.8	61.9	97.9	91.7	97.5	96.7	10/1/0
MSMA*	89.1	94.3	94.7	73.1	100.0	100.0	95.1	99.9	5/5/1
DiffPath	83.8	86.9	88.1	73.5	99.9	100.0	97.1	99.9	6/4/1

Table S28: Chest X-rays (ChestMNIST, 32×32 greyscale) as ID: AUROC (%) of the 1,000 ID test radiographs against their transformed copies, other modalities, chest X-rays of another source and natural images; three noise seeds for Upright kNN, its terms, kNN (basic) and CFS. Held-out rotation accuracy of the head 99.9%. W/T/L of Upright kNN over the 11 sets (bootstrap rule of the paper).

Method	ID transformed				other modality			natural	
	rot. 90	rot. 180	upside down	mirror	blood	derma	CXR	mean (10)	W/T/L
Upright kNN	58.9	59.5	55.3	50.8	87.2	95.9	93.3	97.2	–
kNN term	49.7	49.6	48.8	48.8	95.0	99.4	99.4	99.5	4/0/13
orient. term	66.5	66.9	61.4	53.3	56.1	20.1	0.2	54.7	13/0/4
kNN (basic)	49.4	49.3	48.7	48.7	92.2	99.3	99.6	99.5	4/0/13
CFS (diag.)	50.6	50.1	50.4	50.4	99.9	96.8	99.7	99.0	4/2/11
CFS (kNN)	50.0	49.9	49.3	49.5	100.0	99.9	100.0	100.0	3/1/13
NLL	50.6	50.9	50.8	50.2	89.9	56.9	31.9	91.1	10/5/2
MSMA*	50.4	50.0	50.3	50.3	89.3	83.9	97.9	90.7	14/1/2
DiffPath	52.0	51.0	51.7	50.9	99.8	99.0	99.5	99.6	3/1/13

Table S29: Histology tiles (PathMNIST, 32×32 RGB) as ID, as Table S28; W/T/L over the 17 sets. Held-out rotation accuracy of the head 75.8%: the tiles have no canonical up, and the orientation term, which reads acquisition cues, is inverted on other modalities and lowers the kNN term.

Augmentation	head	Upright kNN		kNN term		orient. term		NLL of the generator		NLL $+ \log P(\mathrm{up} \mid \mathbf{x})$	
	acc.	natural	flip	natural	flip	natural	flip	natural	flip	natural	flip
none	89.0	85.0	91.3	80.9	63.8	84.3	97.7	52.6	51.0	52.7	51.1
horizontal flips	89.2	84.1	90.9	79.5	63.0	84.9	97.9	52.6	51.0	52.6	51.1
rotations, $p = 0.25$	89.2	84.5	90.5	80.9	67.3	83.9	97.7	52.5	50.7	52.6	50.8
rotations, $p = 0.5$	89.1	85.6	90.0	82.0	69.2	83.3	97.4	52.4	50.4	52.4	50.5
rotations, $p = 1$	88.9	84.9	89.9	81.6	62.9	81.7	97.8	52.2	50.0	52.2	50.1

Table S30: CIFAR-10 generators trained by us with the same recipe and different augmentations (none; horizontal flips; a random multiple of 90° with probability p): held-out accuracy (%) of the head trained on the upright bank, and AUROC (%) on the nine natural OOD sets (mean) and on the upside-down probe of Upright kNN, its terms, the generator's own PF-ODE log-likelihood, and the likelihood completed by the orientation term as in Proposition 1 (final evaluation images of the CIFAR-10 tier; three noise seeds; the likelihood once). The frame survives in the features of a generator whose density is rotation-symmetric by construction ($p = 1$), whereas the likelihood is at chance on flips and the completed likelihood too, because the per-image spread of the likelihood (thousands of nats) swamps the frame factor (a few nats).

S17 Compute and Reproducibility

All features were extracted on NVIDIA RTX 2080 Ti GPUs (11 GB). Offline cost of Upright kNN: one pass of the denoiser over the four rotations of the bank (about 7 minutes for 50,000 CIFAR-10 images on one GPU) and about 10 s of head training. Test cost: one denoiser pass, inner products with the bank, and the pooled grid statistics and MLP (about 2 ms on a CPU). The supplementary archive contains the anonymised code (feature extraction, the head, the score, every comparison method as run, and the analysis scripts that read the per-image scores and write every number and table of the paper and of this document), the protocol documents written before each evaluation (`PROTOCOL_UKNN.md` and the others cited above), the per-image scores of every method on the final evaluation images, and the index files of the three splits. The hyper-parameter ranges tried are those stated in Section S1 (20 configurations of the kNN term on the selection data; the head's input set, depth and temperature on held-out bank images); all final values are in Table S1.

References

[Chen *et al.*, 2024] Huanran Chen, Yinpeng Dong, Zhengyi Wang, Xiao Yang, Chengqi Duan, Hang Su, and Jun Zhu. Robust classification via a single diffusion model. In *ICML*, 2024.

[Graham *et al.*, 2023] Mark S. Graham, Walter H. L. Pinaya, Petru-Daniel Tudosiu, Parashkev Nachev, Sebastien Ourselin, and M. Jorge Cardoso. Denoising diffusion models for out-of-distribution detection. In *CVPR Workshops*, 2023.

[Helber *et al.*, 2019] Patrick Helber, Benjamin Bischke, Andreas Dengel, and Damian Borth. EuroSAT: A novel dataset and deep learning benchmark for land use and land cover classification. *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, 12(7):2217–2226, 2019.

[Hinton, 2002] Geoffrey E. Hinton. Training products of experts by minimizing contrastive divergence. *Neural Computation*, 14(8):1771–1800, 2002.

[Kamkari *et al.*, 2024] Hamidreza Kamkari, Brendan Leigh Ross, Jesse C. Cresswell, Anthony L. Caterini, Rahul G. Krishnan, and Gabriel Loaiza-Ganem. A geometric explanation of the likelihood OOD detection paradox. In *ICML*, 2024.

[Karras *et al.*, 2022] Tero Karras, Miika Aittala, Timo Aila, and Samuli Laine. Elucidating the design space of diffusion-based generative models. In *NeurIPS*, 2022.

[Karras *et al.*, 2024] Tero Karras, Miika Aittala, Jaakko Lehtinen, Janne Hellsten, Timo Aila, and Samuli Laine. Analyzing and improving the training dynamics of diffusion models. In *CVPR*, 2024.

[Liu *et al.*, 2023] Zhenzhen Liu, Jin Peng Zhou, Yufan Wang, and Kilian Q. Weinberger. Unsupervised out-of-distribution detection with diffusion inpainting. In *ICML*, 2023.

[Nichol and Dhariwal, 2021] Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *ICML*, 2021.

[Tack *et al.*, 2020] Jihoon Tack, Sangwoo Mo, Jongheon Jeong, and Jinwoo Shin. CSI: Novelty detection via contrastive learning on distributionally shifted instances. In *NeurIPS*, 2020.

[Yang *et al.*, 2023] Jiancheng Yang, Rui Shi, Donglai Wei, Zequan Liu, Lin Zhao, Bilian Ke, Hanspeter Pfister, and Bingbing Ni. MedMNIST v2 – a large-scale lightweight benchmark for 2D and 3D biomedical image classification. *Scientific Data*, 10:41, 2023.